CHAIN RULE

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Name: Solutions

Theorem (The Chain Rule)**.** *Assume that g is differentiable at x and f is differentiable at $g(x)$. Then composition of f with g, $f \circ g(x) = f\left(g(x)\right)$, is differentiable at x and*

$$\frac{\mathrm{d}}{\mathrm{d}x} f \circ g(x) = f'\left(g(x)\right) \cdot g'(x).$$

1. Let $f(x) = (3x^2+1)^2$.

(a) Expand $f(x)$, then take the derivative.

$$f(x) = (3x^2)^2 + 2(3x^2)(1) + (1)^2 = 9x^4 + 6x^2 + 1$$

$$f'(x) = \boxed{36x^3 + 12x}$$

(b) Write $f(x) = (3x^2+1)^2 = (3x^2+1)(3x^2+1)$ and apply the Product Rule.

$$f'(x) = (6x)(3x^2+1) + (3x^2+1)(6x)$$

$$= (3x^2+1)(6x+6x) = 12x(3x^2+1) = \boxed{36x^3+12x}$$

(c) Apply the chain rule directly to $f(x)$.

$$f'(x) = 2(3x^2+1)^{2-1} \frac{d}{dx}(3x^2+1) = 2(3x^2+1)(6x)$$

$$= 12x(3x^2+1) = \boxed{36x^3+12x}$$

(d) Are your answers in parts (a), (b), and (c) the same? Why or why not?

Yes; because all three are valid ways to compute $f'(x)$.

2. Assume that f is a differentiable function and let $g(x) = \left(f\left(\sqrt{x}\right)\right)^3$.

(a) Compute $g'(x)$. Your answer should include both f and f'.

$$g'(x) = 3\left(f(\sqrt{x})\right)^2 \frac{d}{dx}\left(f(\sqrt{x})\right)$$

$$= 3\left(f(\sqrt{x})\right)^2 f'(\sqrt{x}) \frac{d}{dx}\sqrt{x}$$

$$= 3\left(f(\sqrt{x})\right)^2 f'(\sqrt{x})\left(\tfrac{1}{2}x^{-1/2}\right) = \boxed{\frac{3\left(f(\sqrt{x})\right)^2 f'(\sqrt{x})}{2\sqrt{x}}}$$

(b) If $f(2) = 1$ and $f'(2) = -2$, calculate $g'(4)$.

$$g'(4) = \frac{3f(\sqrt{4})^2 f'(\sqrt{4})}{2\sqrt{4}} = \frac{3f(2)^2 f'(2)}{4} = \frac{3(1)^2(-2)}{4} = \boxed{-\frac{3}{2}}$$

Find the derivative of the given function.

3. $f(x) = (1 + x + x^2)^{99}$

$$f'(x) = 99(1+x+x^2)^{98}(0+1+2x)$$

$$= \boxed{99(1+x+x^2)^{98}(1+2x).}$$

4. $g(\theta) = (2 - \sin(\theta))^{3/2}$

$$g'(\theta) = \frac{3}{2}(2-\sin(\theta))^{1/2}(0-\cos(\theta))$$

$$= \boxed{-\frac{3}{2}(2-\sin(\theta))^{1/2}\cos(\theta).}$$

5. $f(x) = (2x-3)^4(x^2+x+1)^5$.

$$f'(x) = 4(2x-3)^3(2)(x^2+x+1)^5 + (2x-3)^4(5)(x^2+x+1)^4(2x+1)$$

$$= \boxed{8(2x-3)^3(x^2+x+1)^5 + 5(2x+1)(2x-3)^4(x^2+x+1)^4}$$

6. $g(\theta) = \cos(\theta)^2$

$$g'(\theta) = 2\cos(\theta)(-\sin(\theta)) = \boxed{-2\cos(\theta)\sin(\theta).}$$

7. $h(y) = \left(\dfrac{y^4+1}{y^2+1}\right)^5$

$$h'(y) = 5\left(\frac{y^4+1}{y^2+1}\right)^4\left(\frac{4y^3(y^2+1)-(y^4+1)(2y)}{(y^2+1)^2}\right) = \frac{5(y^4+1)^4(4y^5+4y^3-2y^5-2y)}{(y^2+1)^6}$$

$$= \boxed{\frac{5(y^4+1)^4(2y^5+4y^3-2y)}{(y^2+1)^6}}$$

8. $f(\theta) = \cot^2(\sin(\theta))$

$$f'(\theta) = 2\cot(\sin(\theta))\frac{d}{d\theta}\cot(\sin(\theta))$$

$$= 2\cot(\sin(\theta))(-\csc^2(\sin(\theta)))\frac{d}{d\theta}\sin(\theta)$$

$$= \boxed{-2\cot(\sin(\theta))\csc^2(\sin(\theta))\cos(\theta)}$$

9. $f(t) = \sqrt{\dfrac{t}{t^2+4}} = \left(\dfrac{t}{t^2+4}\right)^{1/2}$

$$f'(t) = \frac{1}{2}\left(\frac{t}{t^2+4}\right)^{-1/2}\left(\frac{t^2+4-t(2t)}{(t^2+4)^2}\right) = \frac{t^{-1/2}(-t^2+4)}{2(t^2+4)^{-1/2}(t^2+4)^2} = \frac{-t^2+4}{2\sqrt{t}(t^2+4)^{2-1/2}}$$

$$= \boxed{\frac{-t^2+4}{2\sqrt{t}(t^2+4)^{3/2}}}$$